Synopsis:

“BEHAVIORAL SEGMENTATION FOR TARGETED MARKETING”

Problem Id: 10

Student Name: Akshita Bahuguna(2028387)

Student Name: Avika Dixit(2028404)

Student Name:Vanya Rathi(2028514)

Section : L

Problem Description

In the current competitive retail environment, it is essential for businesses to understand the behavior of customers to enhance their marketing strategies and improve profitability. The current project is based on behavioral segmentation, in which customers are divided into different segments according to their behavior in terms of spending and frequency of purchase. Based on these characteristics, businesses can identify different personas and target those personas with appropriate marketing strategies.

In the current project, a synthetic data set of 150 customers will be generated and machine learning techniques will be implemented to find the patterns. StandardScaler will be implemented for normalizing the features and K-Means clustering for segmenting the customers into appropriate clusters. This will help businesses improve the effectiveness of the marketing strategy and increase the efficiency of the sales process.

Steps of Implementation:

1.Data Collection:

First, we will gather a dataset that includes parameters like name,age,gender,city,annual spending ,etc.

Feature Name	**Description**	**Data Type**	**Possible Values**
id	Unique identifier for each customer	Integer	1, 2, 3, …
age	Age of the customer	Integer	18 – 70
gender	Gender of the customer	Categorical	Male, Female, Other
income	Annual income of the customer	Integer	20,000 – 150,000

spending	Spending score indicating customer behavior	Integer	1 – 100
members	Number of family members	Integer	1 – 10
purchase_frequency	Number of purchases made in a period	Integer	1 – 50
preferred_category	Preferred shopping category	Categorical	Groceries, Clothing, Electronics, Sports, Home & Garden
last_purchase_amount	Amount spent in the last purchase	Float	10.0 – 1000.0

2. Data Preprocessing:

Next, we will clean the data by handling missing values and removing unwanted noise. We will also scale the data so that all values are in a similar range, which helps the model perform better.

To improve model performance, the following preprocessing steps will be applied:

- Handling Missing Values
  - Use mean/median imputation for numerical data
  - Use mode for categorical data
- Noise Removal
  - Remove outliers using IQR (Interquartile Range) or Z-score method

    Z = (x-u) / o (u is mean, o is standard deviation)

- Encoding Categorical Data
  - Convert Gender or City categories using Label Encoding / One-Hot Encoding
- Feature Selection
  - Use correlation matrix or feature importance to remove less useful features

    Min -Max Scaling : Xnorm = (X-Xmin) / (Xmax-Xmin)

    X-Standard : Xstd = (X-u) / o (u is mean, o is standard deviation)

3. Model Selection:

We will use the following algorithms:

K-Means Clustering

K-means clustering is a popular unsupervised machine learning algorithm used for partitioning a dataset into a pre-defined number of clusters. The goal is to group similar data points together and discover underlying patterns or structures within the data.
Recall the first property of clusters – it states that the points within a cluster should be similar to each other. So, our aim here is to minimize the distance between the points within a cluster.
There is an algorithm that tries to minimize the distance of the points in a cluster with their centroid – the k-means clustering technique.
K-means is a centroid-based algorithm or a distance-based algorithm, where we calculate the distances to assign a point to a cluster. In K-Means, each cluster is associated with a centroid.

4. Training and Validation:

- The dataset will be split into:
  - Training Set (70–80%)
  - Testing Set (20–30%)
- Training Process:
  - Models (K- Means Clustering) will be trained using training data
- Validation Techniques:
  - Train-Test Split Method
  - Cross-Validation for better reliability

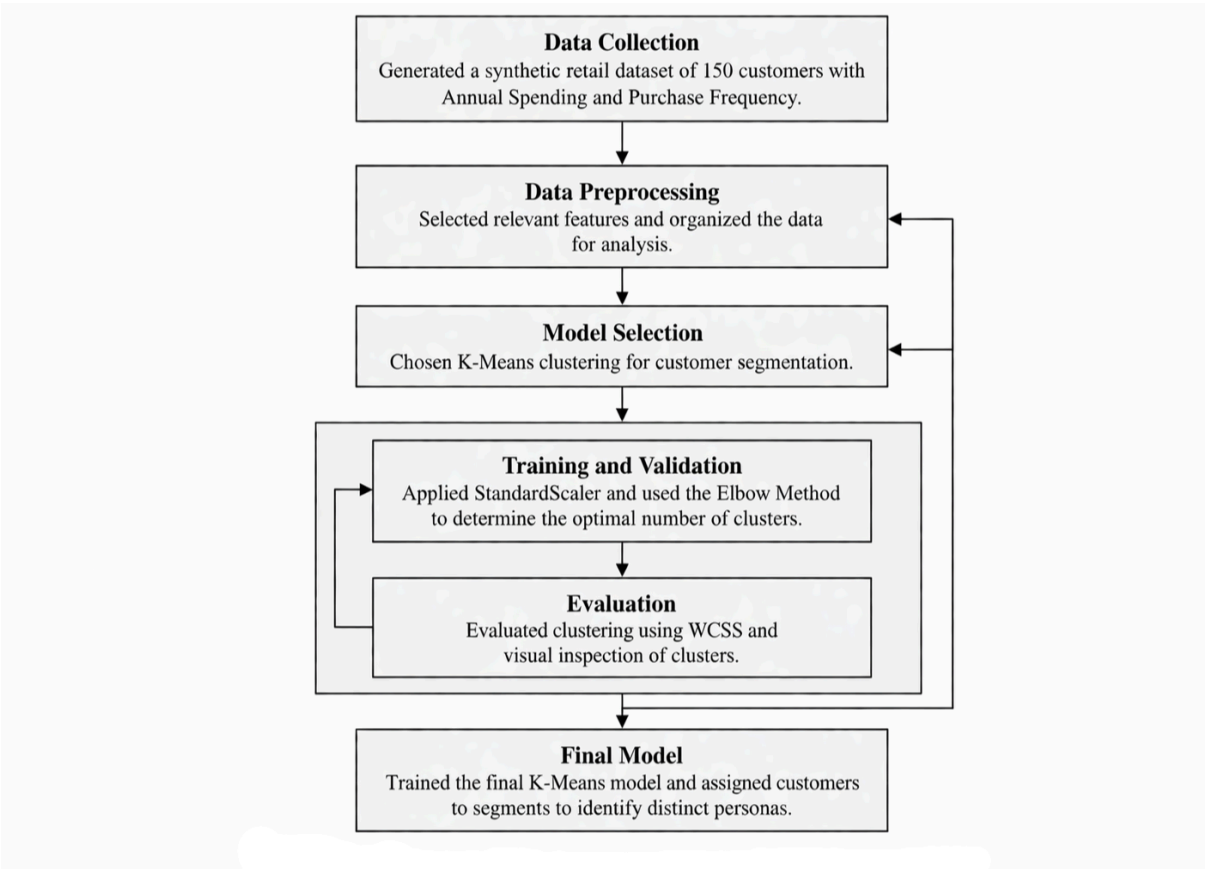
5. Evaluation:

The performance of models will be evaluated using:

- Accuracy : Overall correctness
- Precision : Correct positive predictions
- Confusion Matrix : Detailed classification results

6. Final Model:

Finally, we will select the model that gives the best results and use it for predicting air quality levels.



Expected Modules/ Libraries to be used :

NumPy:

NumPy is a very important library that is used in project for handling numbers quickly and easily. It does the work with big lists of numbers (arrays and matrices), which is really helpful when dealing with a lot of customer data. NumPy makes it simple to do math like finding the average, variance, and standard deviation much faster than using normal Python loops. In this project, NumPy is used to generate random datasets, modify the data during preprocessing, and assist with scaling and normalisation. Since NumPy can do calculations on whole arrays at once, it saves a lot of time and makes working with data much more efficient.

Pandas:

Pandas is a data analysis and manipulation library that plays a central role in organising and processing datasets. It introduces the DataFrame structure, which allows data to be stored in a tabular format with labelled rows and columns, making it easy to work with structured customer information such as age, gender, spending, and purchase frequency. In this project, Pandas is used extensively for data cleaning tasks such as handling missing values (using mean, median, or mode imputation), removing duplicate entries, and transforming raw data into a usable format. It also supports advanced operations like grouping, aggregation, and filtering, which help in analyzing customer behaviour patterns. Additionally, it simplifies the process of encoding categorical variables and exporting/importing datasets in formats like CSV.

Matplotlib:

Matplotlib is a comprehensive data visualization library used to represent data graphically, which is important for understanding patterns and validating results in the project. It enables the creation of various types of plots such as scatter plots, line graphs, bar charts, and specifically the elbow curve used in clustering analysis. In this project, scatter plots help visualize how customers are grouped based on features like annual spending and purchase frequency, providing intuitive insights into different customer segments. The elbow method graph, created using Matplotlib, helps determine the optimal number of clusters by plotting the Within-Cluster Sum of Squares (WCSS) against different values of K. These visual tools not only enhance interpretability but also support decision-making by clearly presenting patterns and relationships in the data.

Scikit-learn:

Scikit-learn is a widely used machine learning library that provides efficient and easy-to-implement tools for data preprocessing, model building, and evaluation. In this project, it is primarily used for implementing StandardScaler, which standardizes the dataset by transforming features to have zero mean and unit variance—an essential step for distance-based algorithms like K-Means. The K-Means clustering algorithm from Scikit-learn is used to segment customers into different groups based on behavioral similarities such as spending habits and purchase frequency. The library also provides utilities for splitting data into training and testing sets, as well as evaluation techniques to validate model performance.

Expected Evaluation metrics and Plots to be used:

Elbow Method (WCSS):

Used to determine optimal number of clusters by minimizing within-cluster variance.

Scatter Plot:

Helps visualize customer segments based on spending and frequency.

Cluster Visualization:

Different colors represent different customer groups for easy interpretation.

Required topics from the subject:

Data Structures (Arrays & DataFrames):

Used to store and manage customer data like Annual Spending and Purchase Frequency using NumPy arrays and Pandas DataFrames.

Functions:

Used to structure the program into reusable blocks such as data generation, scaling, and clustering steps.

File Handling:

Used to save and load the dataset (e.g., CSV files) for future use and analysis.

Loops & Conditionals:

Used in implementing the Elbow Method (loop over different cluster values) and making decisions in the program.

Libraries & Modules:

Used in implementing the Elbow Method (loop over different cluster values) and making decisions in the program.

Platform Required:

Jupyter Notebook:

Used as the primary development environment for writing, executing, and interactively visualising the Python Code.

Python (Anaconda Distribution):

Provides all required libraries and package management needed for data analysis and machine learning.

Books and Link Sources:

1. "Hands-On Machine Learning with Scikit-Learn" by Aurélien Géron
2. Python for Data Analysis" by Wes McKinney
3. Scikit-learn Documentation (https://scikit-learn.org/)
4. Kaggle (https://www.kaggle.com/)